[INCH-POUND]
A-A-59550/1A
March 14, 2005
SUPERSEDING
A-A-59550/1
May 17, 2000

COMMERCIAL ITEM DESCRIPTION

FUSE, INCLOSED LINK, THIN-FILM, SURFACE MOUNT (SM), 0603 SIZE, VERY FAST ACTING

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the issue in effect of CID A-A-59550.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59550	/01	-	001
CID number	Applicable CID sheet (two digits)		Dash number (see applicable CID sheet)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Fuses supplied to this CID shall be as specified herein (see figure 1).

Electrical specifications.

Ampere rating. The ampere rating shall be as specified in table I.

Voltage rating. The voltage rating shall be 32 V ac/dc.

Interrupting ratings. The interrupting rating shall be 50 amperes at 32 Vac/dc for ampere ratings of 0.25 ampere through 1 ampere and 35 amperes at 32 Vac/dc for ampere ratings of 1.5 amperes through 5 amperes.

Opening time characteristics. The opening time characteristics shall be as indicated in table II.

Environmental specifications. Fuses supplied to this CID shall be subject to the following tests and there shall be no electrical or mechanical damage to the fuse.

Operating temperature. The operating temperature shall be -55°C to +125°C.

Vibration. Fuses shall withstand 10 Hz - 55 Hz in accordance with method 201A, MIL-STD-202 and 10 Hz – 2000 Hz at 20 G's in accordance with method 204D, condition D of MIL-STD-202.

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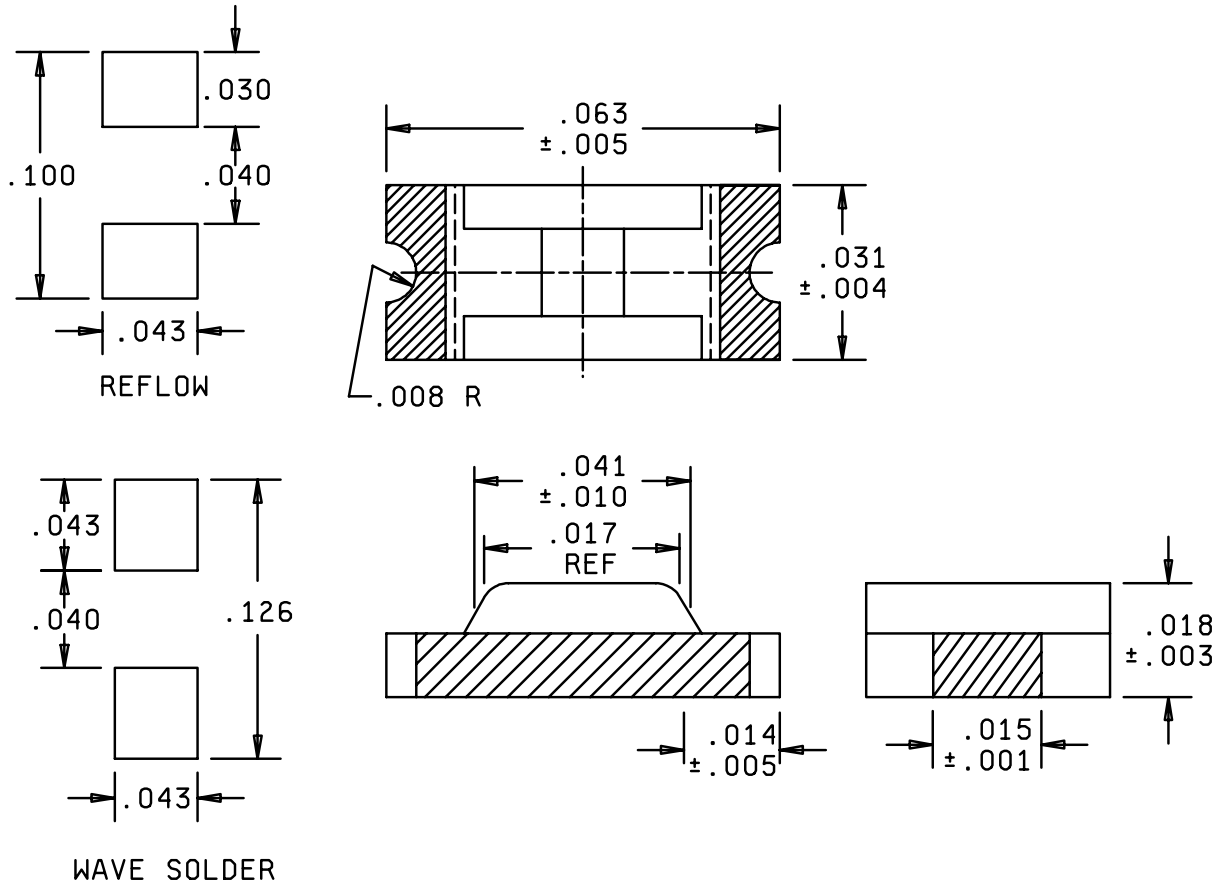

Inches	mm	Inches	mm
.001	0.03	.018	0.46
.003	0.08	.030	0.76
.004	0.10	.031	0.79
.005	0.13	.040	1.02
.008	0.20	.041	1.04
.010	0.25	.043	1.09
.014	0.36	.063	1.60
.015	0.38	.100	2.54
.017	0.43	.126	3.20

NOTES:

1. Dimensions are in inches.
2. Tolerance is ±.005 inch(0.13 mm), unless otherwise specified.

FIGURE 1. Configuration and dimensions.

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TABLE I. Electrical characteristics.

AA59450/01-	Ampere rating	Nominal resistance cold ohm 1/	Nominal voltage drop (V) 2/
001	1/4	0.340	0.130
002	3/8	0.218	0.084
003	1/2	0.170	0.120
004	3/4	0.100	0.110
005	1	0.072	0.100
006	1 1/2	0.049	0.082
007	2	0.036	0.077
008	2 1/2	0.028	0.089
009	3	0.023	0.080
010	3 1/2	0.019	0.090
011	4	0.017	0.100
012	5	0.013	0.107

1/ Measured at 10 percent of rated current, 25°C.
2/ Measured at 100 percent of rated current, 25°C.

TABLE II. Opening time characteristics.

Percent of ampere rating	Opening time at 25°C
100 percent	4 hours, minimum
200 percent	5 seconds, maximum
300 percent	0.2 seconds, maximum

Insulation resistance (after opening). The insulation resistance after opening shall be 500,000 ohms minimum.

Resistance to soldering heat. Fuses shall be able to withstand 60 seconds above 200°C up to 260°C , maximum.

Thermal shock. Fuses shall be able to withstand 5 cycles of -55°C to +125°C.

Physical specifications.

Materials. Fuses shall have an epoxy substrate body, Copper/Nickel/Tin-lead (95/5), cover coat of conformal coating.

Soldering parameters. Fuses shall be able to withstand, without electrical or mechanical damage to the fuse, a wave solder of +260°C (+500°F) for 10 seconds maximum, and a reflow solder of +260°C (+500°F) for 30 seconds maximum.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

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Source of documents.

Military standards

MIL-STD-202 - Test Method Standard Electronic and Electrical Component Parts.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://dodssp.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
75915	Littelfuse Incorporated 800 E. Northwest Highway Des Plaines, IL 60016-3096 Phone number (847) 824-1188 Facsimile number (847) 391-0894 E-mail mailto:electronics@littelfuse.com Uniform Resource Locator (URL) http://www.littelfuse.com

Part number (P/N) supersession data. These CID PINs supersede the following MFR's P/N's as shown. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

CID dash number (see table I)	Vendor commercial P/N 1/	
AA59550/01-	MFR's CAGE	MFR's P/N 1/
001	75915	431.250
002	75915	431.375
003	75915	431.500
004	75915	431.750
005	75915	431001
006	75915	43101.5
007	75915	431002
008	75915	43102.5
009	75915	431003
010	75915	43103.5
011	75915	431004
012	75915	431005

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown. For actual part marking requirements see the marking paragraph.

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MILITARY INTERESTS:

Custodians:
Navy - EC
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

DLA-CC

Project 5920-0858-01

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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